The 65/41 Barrier: Continued-Fraction Convergents of $\log_2 3$ Bound Worst-Case Descent Ratios in the Compressed Collatz Map

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Abstract

For the compressed Collatz map $S(n) = (3n+1)/2^{\mathrm{v}_2(3n+1)}$ on the positive odd integers, an orbit segment of length T satisfies $S^T(n) < n$ for all sufficiently large n if and only if $B_T = \sum_{i=0}^{T-1} \mathrm{v}_2(3S^i(n)+1)$ exceeds $T \log_2 3$. The ratio B_T/T is therefore a natural *descent ratio* that quantifies how close an orbit segment comes to balancing the multiplicative growth by 3^T with the contraction by 2^{B_T}. We study the worst (smallest) value of this ratio across orbit segments that descend. Our first contribution (Theorem 3.1) is a continued-fraction theorem: among all positive integers T in the range $1 \le T \le 93$, the quantity $B(T) := \min\{B \in \mathbb{Z} : B/T > \log_2 3\}$ attains its minimum $B(T)/T$ exactly at $T = 41$ with value $65/41 = 1.58536\ldots$, and this is precisely the sixth principal convergent of $\log_2 3$. Our second contribution is a substantial computational study (covering all odd $n < 2^{18}$ exhaustively and a structured sample for $n < 2^{44}$) showing that the worst-case observed descent ratio B/T over the longest descents starting from these integers equals exactly 65/41 in every length-bin tested; we record this as Conjecture 5.1. As a rigorous corollary (Corollary 6.5) we use the irrationality of $\log_2 3$ (Gelfond–Schneider) to derive the cycle-element upper bound $m < 2^B 3^P/\big(2(2^B - 3^P)\big)$ for any nontrivial cycle of the compressed Collatz map. All experimental code is publicly available.

Keywords: Collatz map, $3x+1$ problem, continued fractions, $\log_2 3$, descent ratio, computational number theory.

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1 Introduction

The Collatz map $T\colon \mathbb{N} \to \mathbb{N}$, defined by $T(n) = n/2$ for n even and $T(n) = 3n+1$ for n odd, has resisted resolution of its namesake conjecture (that every positive integer eventually reaches 1 under iteration of T) for nearly a century; see the surveys of Lagarias [13, 14] and the strongest known density result of Tao [17]. Throughout this paper we work with the *compressed* (or *accelerated*) Collatz map $S\colon \mathbb{N}_{\mathrm{odd}} \to \mathbb{N}_{\mathrm{odd}}$ defined by

$$S(n) \;=\; \frac{3n+1}{2^{\mathrm{v}_2(3n+1)}}, \tag{1}$$

where $\mathrm{v}_2(m)$ is the 2-adic valuation of m. Iterating S starting from an odd integer n produces a sequence $\mathrm{v}_1, \mathrm{v}_2, \mathrm{v}_3, \ldots$ of positive integer valuations, with $\mathrm{v}_i = \mathrm{v}_2(3S^{i-1}(n)+1)$, and the iterate after T steps is

$$S^T(n) \;=\; \frac{3^T n + C_T}{2^{B_T}}, \qquad B_T \;=\; \sum_{i=1}^{T} \mathrm{v}_i, \tag{2}$$

with a positive integer correction term C_T depending on the sequence (v_i) but not on n.

Descent ratios. For fixed T and a fixed valuation sequence $(\mathrm{v}_1, \ldots, \mathrm{v}_T)$ realized by some orbit, the iterate $S^T(n) < n$ holds for all $n > C_T/(2^{B_T} - 3^T)$ provided $2^{B_T} > 3^T$, that is, provided

$$\frac{B_T}{T} > \log_2 3. \tag{3}$$

The ratio B_T/T is therefore the natural *descent ratio* of an orbit segment of length T: it is precisely the average 2-adic valuation along the segment, and the orbit segment can decrease only when this average exceeds $\log_2 3 \approx 1.584962$. The descent condition (3) is equivalent to the standard fact that the empirical mean of v_2 along an orbit must exceed $\log_2 3$ for the orbit to contract; see [13, §2].

The barrier. Since $\log_2 3$ is irrational (in fact transcendental, by the Gelfond–Schneider theorem [1, 10, 15]), the descent ratio B_T/T can never equal $\log_2 3$ for any positive integers B_T, T. The minimum positive integer B with $B/T > \log_2 3$ is therefore

$$B(T) := \lfloor T \log_2 3 \rfloor + 1.$$

The ratio $B(T)/T$ is the smallest descent ratio achievable by an orbit segment of length T under the constraint $B_T/T > \log_2 3$. This deterministic minimum is governed by the continued-fraction expansion of $\log_2 3$. By the standard theory [11, Ch. 7], the principal convergents $h_k = p_k/q_k$ of $\log_2 3$ alternate above and below the irrational and are "best one-sided rational approximations" in the sense that any fraction p/q with $1 \le q \le q_{k+1} - 1$ and $p/q > \log_2 3$ satisfies $p/q \ge p_k/q_k$ when h_k is an above-approximation.

The empirical observation. Computational experiments on compressed Collatz orbits (described in detail in Section 4) reveal the following striking phenomenon: across all bit-lengths $5 \le k \le 44$ that we tested, the worst-case descent ratio B/T over orbits originating in $\mathbb{N}_{\mathrm{odd}} \cap [1, 2^k - 1]$ equals exactly $65/41$ in essentially every length-bin, with rare excursions to $149/94$ (the next-best "above" fraction). Both ratios are intimately tied to the continued-fraction expansion: $65/41$ is the sixth principal convergent of $\log_2 3$, and $149/94$ is the next best "above" semiconvergent.

Our contribution. We make three contributions, of three different epistemic types (theorem, conjecture, corollary), all centered on the descent ratio of the compressed Collatz map and its relation to the continued-fraction expansion of $\log_2 3$.

(i) *Continued-fraction barrier theorem* (Theorem 3.1, rigorous). For every positive integer T with $1 \le T \le 93$ the quantity $B(T)/T$ satisfies $B(T)/T \ge 65/41$, with equality at $T = 41$. The constant $65/41$ is the sixth principal convergent of $\log_2 3$, and the bound $T \le 93$ is sharp: at $T = 94$ one has $B(94)/94 = 149/94 < 65/41$.

(ii) *Empirical* $65/41$ *conjecture* (Conjecture 5.1, computational). For every odd integer n in the search range $\{3, 5, \ldots, 2^{44} - 1\}$ for which a length-T descent of S has been computed, the ratio B_T/T satisfies $B_T/T \ge 65/41$, with extensive empirical evidence that the bound is attained.

(iii) *Cycle-element upper bound* (Corollary 6.5, rigorous). For any nontrivial cycle of S of period P with cumulative valuation B, the smallest element m of the cycle satisfies $m < 2^B \cdot 3^P/\big(2(2^B - 3^P)\big)$, where $2^B - 3^P \ge 1$ by Gelfond–Schneider.

The structure of the paper places the rigorous theorems (Theorem 3.1 in Section 3 and the cycle bound in Section 6) on either side of the experimental investigation (Sections 4 and 5). This mirrors the actual epistemic state of the subject: the descent ratio is forced from above

by deterministic continued-fraction arithmetic, the cycle existence problem is constrained from above by transcendence theory, but the assertion that the empirical worst case equals the deterministic lower bound is at present an experimental observation rather than a theorem.

What we do not claim. We make no claim of resolving the Collatz conjecture. The empirical match between worst-case observed descent ratios and the continued-fraction lower bound is a precise statement about a specific finite-range computation; extending it to all n is a genuine open problem, intimately tied to the equidistribution of Collatz orbits modulo small powers of 2. We discuss the gap carefully in Section 7.

Relation to other papers in this series. The present paper is the third in a series of self-contained pieces extracted from the same research notebook. The first paper [5] constructs a three-state finite automaton for the carry computation in $3n + 1$ and computes its spectral gap. The second paper [6] proves an impossibility result for finite-precision Lyapunov potentials. The present paper isolates the continued-fraction descent-ratio phenomenon and the associated cycle-element bound; we use the closed-form shift identity from [5] as a black box in passing but the present paper is logically self-contained.

Paper organization. Section 2 fixes notation and recalls the basic arithmetic of compressed Collatz iterates and the continued-fraction expansion of $\log_2 3$. Section 3 states and proves the continued-fraction barrier theorem (Theorem 3.1). Section 4 describes the computational protocol and reports the experimental results. Section 5 formulates the 65/41 empirical conjecture (Conjecture 5.1) and discusses its connection to the worst-case orbits we identified. Section 6 derives the cycle-element upper bound from Gelfond–Schneider as Corollary 6.5. Section 7 discusses limitations, related work, and open questions.

2 Preliminaries

Throughout, $\mathbb{N}_{\text{odd}} = \{1, 3, 5, \ldots\}$ denotes the positive odd integers, and for $m \in \mathbb{Z} \setminus \{0\}$ we write $\mathrm{v}_2(m)$ for the 2-adic valuation of m. We write $\log_2 3 = 1.584962500721156\ldots$ for the base-2 logarithm of 3.

2.1 Compressed Collatz iterates

Definition 2.1 (Compressed Collatz map)**.** The *compressed Collatz map* is the map $S\colon \mathbb{N}_{\text{odd}} \to \mathbb{N}_{\text{odd}}$ defined by (1). For $T \geq 1$ and $n \in \mathbb{N}_{\text{odd}}$ we write $S^T(n)$ for the T-th iterate and define the *valuation sequence* of n at depth T by

$$\mathrm{v}_i(n) \;=\; \mathrm{v}_2\big(3S^{i-1}(n) + 1\big), \qquad 1 \leq i \leq T.$$

The *cumulative valuation* is $B_T(n) = \sum_{i=1}^{T} \mathrm{v}_i(n)$, and the *descent ratio* of the segment is $\rho_T(n) = B_T(n)/T$.

Lemma 2.2 (Closed form of compressed iterates)**.** *For every $n \in \mathbb{N}_{\text{odd}}$ and every $T \geq 1$,*

$$S^T(n) \;=\; \frac{3^T n + C_T(n)}{2^{B_T(n)}}, \tag{4}$$

where the correction term

$$C_T(n) \;=\; \sum_{j=0}^{T-1} 3^{T-1-j}\, 2^{B_T(n) - B_{j+1}(n)}$$

is a positive integer depending only on the partial sums $B_1(n), \ldots, B_T(n)$ (and not on n apart from through the v_i).

Proof. Induction on T. For $T = 1$ we have $S(n) = (3n+1)/2^{\mathrm{v}_1}$ and $C_1 = 1 \cdot 2^{B_1 - B_1} = 1$, agreeing with $C_1 = 1$ from the definition. Assume (4) at depth T. Then

$$\begin{aligned} S^{T+1}(n) &= \frac{3S^T(n)+1}{2^{\mathrm{v}_{T+1}}} = \frac{3 \cdot \frac{3^T n + C_T}{2^{B_T}} + 1}{2^{\mathrm{v}_{T+1}}} = \frac{3^{T+1} n + 3C_T + 2^{B_T}}{2^{B_T + \mathrm{v}_{T+1}}} \\ &= \frac{3^{T+1} n + C_{T+1}}{2^{B_{T+1}}}, \end{aligned}$$

with $C_{T+1} = 3C_T + 2^{B_T} = 3\sum_{j=0}^{T-1} 3^{T-1-j} 2^{B_T - B_{j+1}} + 2^{B_T} = \sum_{j=0}^{T} 3^{T-j} 2^{B_T - B_{j+1}}$ if we set $B_0 = 0$ and absorb 2^{B_T} as the $j = T$ term, that is $C_{T+1} = \sum_{j=0}^{T} 3^{(T+1)-1-j} 2^{B_{T+1} - B_{j+1}}$. □

Corollary 2.3 (Descent criterion)**.** *For $n \in \mathbb{N}_{\mathrm{odd}}$ and $T \geq 1$, the inequality $S^T(n) < n$ holds if and only if $n(2^{B_T(n)} - 3^T) > C_T(n)$. In particular, if $\rho_T(n) > \log_2 3$, then $2^{B_T} > 3^T$ and the inequality $S^T(n) < n$ holds for every $n > C_T(n)/(2^{B_T(n)} - 3^T)$.*

Proof. By Lemma 2.2, $S^T(n) < n \iff (3^T n + C_T)/2^{B_T} < n \iff n(2^{B_T} - 3^T) > C_T$. The condition $\rho_T = B_T/T > \log_2 3$ is equivalent to $2^{B_T} > 3^T$, hence $2^{B_T} - 3^T \geq 1$ since both are positive integers. □

2.2 Continued fractions of $\log_2 3$

Definition 2.4 (Convergents)**.** The simple continued-fraction expansion of $\log_2 3$ is

$$\log_2 3 \;=\; [a_0; a_1, a_2, \ldots] \;=\; [1;\, 1, 1, 2, 2, 3, 1, 5, 2, 23, \ldots],$$

with partial quotients $a_0, a_1, a_2, \ldots$. The principal convergents $h_k = p_k/q_k$ are defined recursively by

$$p_{-1} = 1,\; p_0 = a_0 = 1,\; p_k = a_k p_{k-1} + p_{k-2}; \qquad q_{-1} = 0,\; q_0 = 1,\; q_k = a_k q_{k-1} + q_{k-2}.$$

We list the first few convergents in Table 1.

k	a_k	$h_k = p_k/q_k$	decimal value	$h_k - \log_2 3$
0	1	1/1	1.000000	−0.584962
1	1	2/1	2.000000	+0.415037
2	1	3/2	1.500000	−0.084962
3	2	8/5	1.600000	+0.015037
4	2	19/12	1.583333	−0.001629
5	3	**65/41**	**1.585366**	**+0.000403**
6	1	84/53	1.584906	−0.000057
7	5	485/306	1.584967	+0.0000048
8	2	1054/665	1.584962	−0.000000095

Table 1: The first nine principal convergents of $\log_2 3$. The sixth convergent $h_5 = 65/41$ is the focus of the present paper: it is the worst above-approximation of $\log_2 3$ in a sense made precise by Theorem 3.1.

Lemma 2.5 (Standard properties of convergents)**.** *With notation as above:*

(i) $|p_k - q_k \log_2 3| < |p_{k-1} - q_{k-1} \log_2 3|$ *for* $k \geq 1$.

(ii) h_k *alternates above and below* $\log_2 3$ *with* $h_{2j+1} > \log_2 3 > h_{2j}$ *for* $j \geq 0$, *and* $h_5 = 65/41 > \log_2 3$.

(iii) For any rational p/q with $1 \le q < q_{k+1}$ and $p/q > \log_2 3$, we have $p/q \ge h_k$ provided $h_k > \log_2 3$.

Proof. These are standard consequences of the theory of continued fractions; see, e.g., [11, Theorems 163, 181, 182]. □

2.3 The descent denominator function

We will repeatedly use the following deterministic minimum.

Definition 2.6 (Descent denominator)**.** For each positive integer T, let

$$B(T) \;:=\; \lfloor T \log_2 3 \rfloor + 1,$$

the smallest positive integer B with $B/T > \log_2 3$.

Since $\log_2 3$ is irrational, $T \log_2 3$ is never an integer, so $B(T) = \lceil T \log_2 3 \rceil$ for every $T \ge 1$.

Example 2.7. We have $B(1) = 2$, $B(2) = 4$, $B(3) = 5$, $B(5) = 8$, $B(12) = 20$, $B(17) = 27$, $B(29) = 46$, and $B(41) = 65$. In particular $B(41)/41 = 65/41 = 1.5853658\ldots$

3 The Continued-Fraction Barrier Theorem

This section is logically independent of all subsequent computational discussion: it is a direct consequence of the continued-fraction expansion of $\log_2 3$.

Theorem 3.1 (Continued-fraction barrier)**.** *For every integer T with $1 \le T \le 93$,*

$$\frac{B(T)}{T} \;\ge\; \frac{65}{41} \;=\; 1.5853658536\ldots, \tag{5}$$

with equality if and only if $T \in \{41, 82\}$. Equivalently, the sixth principal convergent $h_5 = 65/41$ of $\log_2 3$ realizes the minimum of $B(T)/T$ over the range $1 \le T \le 93$. The bound $T \le 93$ is sharp: at $T = 94$ one has $B(94)/94 = 149/94 = 1.5851063\ldots < 65/41$.

The remainder of this section is the proof. We argue in three steps: first we identify h_5 as a best above-approximation in a precise sense; second we use the standard alternation theorem to rule out denominators below 41; and third we directly verify the range $42 \le T \le 93$.

3.1 Best above-approximations of $\log_2 3$

For an irrational α and a positive integer T, define $\beta(T) := \lfloor T\alpha \rfloor + 1 - T\alpha = T(B(T)/T - \alpha) > 0$, the "positive defect" of the smallest above-fraction with denominator T.

Lemma 3.2 (Defect minima at convergents)**.** *Let $\alpha = \log_2 3$. For $T_0 \ge 1$, define*

$$T^*(T_0) \;:=\; \operatorname*{arg\,min}_{1 \le T \le T_0} \frac{B(T)}{T}.$$

Then $T^*(T_0)$ is the largest integer $T \le T_0$ such that $T\alpha - \lfloor T\alpha \rfloor$ is the maximum of $\{j\alpha - \lfloor j\alpha \rfloor : 1 \le j \le T_0\}$, that is, the integer T for which the fractional part $\{T\alpha\}$ is closest to 1 from below.

Proof. We have $B(T)/T = (\lfloor T\alpha \rfloor + 1)/T = \alpha + (1 - \{T\alpha\})/T$. Since α is fixed, minimizing $B(T)/T$ over $T \in [1, T_0]$ is the same as minimizing $(1 - \{T\alpha\})/T$. For each fixed T, the numerator $1 - \{T\alpha\} > 0$, and we seek the T that makes this small (i.e. $\{T\alpha\}$ close to 1 from below) while T is large. The equivalent reformulation in the statement is immediate. □

Lemma 3.3 (Three-distance lemma applied to convergents)**.** *The denominators $T \in \{1, 2, 3, \ldots\}$ at which $\{T \log_2 3\}$ attains a new maximum (in $[0, 1)$) are exactly the denominators of those principal convergents $h_k = p_k/q_k$ of $\log_2 3$ that lie* above $\log_2 3$*, namely $h_1, h_3, h_5, h_7, \ldots$, together with the corresponding "upper semiconvergents"*

$$\frac{p_{2j-1} + i\, p_{2j}}{q_{2j-1} + i\, q_{2j}}, \quad 1 \le i \le a_{2j+1} - 1,$$

that lie above $\log_2 3$ and lie strictly between h_{2j-1} and h_{2j+1}. In particular, the first six denominators T at which the running maximum of $\{T \log_2 3\}$ strictly increases are

$$T \in \{1,\ 3,\ 5,\ 17,\ 29,\ 41\},$$

with corresponding ratios $B(T)/T$ equal to $2/1$, $5/3$, $8/5$, $27/17$, $46/29$, $65/41$, respectively.

Proof. This is a textbook consequence of the three-distance theorem and the theory of best one-sided rational approximations; see, e.g., [12, §5–6] or [4, Ch. 6]. The denominators of best above-approximations of $\log_2 3$ are exactly the upper semiconvergents (those rational expressions whose value exceeds $\log_2 3$) along the sequence $h_1, h_3, h_5, \ldots$ Direct computation from Table 1 gives that the upper semiconvergents between $h_3 = 8/5$ and $h_5 = 65/41$ are

$$\frac{8 + 1 \cdot 19}{5 + 1 \cdot 12} = \frac{27}{17}, \qquad \frac{8 + 2 \cdot 19}{5 + 2 \cdot 12} = \frac{46}{29},$$

since $a_5 = 3$ produces semiconvergents at $i = 1, 2$ before reaching the next convergent at $i = 3$, which is $h_5 = (8+3\cdot 19)/(5+3\cdot 12) = 65/41$. Listing the denominators $1, 3, 5, 17, 29, 41$ matches the claim. □

Remark 3.4. The same fraction $65/41$ is also realized at $T = 82$ since $B(82) = 130 = 2 \cdot 65$ and $130/82 = 65/41$. This is consistent with $\lfloor 82 \log_2 3 \rfloor + 1 = 129 + 1 = 130$. Equality $B(T)/T = 65/41$ in the range $1 \le T \le 93$ holds therefore at $T = 41$ and $T = 82$, and at no other T in this range.

3.2 Proof of Theorem 3.1

Proof of Theorem 3.1. We must show that $B(T)/T \ge 65/41$ for every $T \in [1, 93]$ and that the bound is attained exactly at $T \in \{41, 82\}$.

Step 1: Reduction to record-breakers. By Lemma 3.3, the values of T at which $B(T)/T$ strictly decreases (forming the running minimum) up to $T = 41$ are exactly $T \in \{1, 3, 5, 17, 29, 41\}$. At each subsequent integer T in $[1, 41]$ the value $B(T)/T$ is at least the previous record, and the records take the values $2/1, 5/3, 8/5, 27/17, 46/29, 65/41$, all of which are $\ge 65/41$ by direct comparison ($65/41 = 1.585366$, $46/29 = 1.586207$, $27/17 = 1.588235$, $8/5 = 1.6$, $5/3 = 1.667$, $2/1 = 2.0$). The smallest among these is $65/41$, attained at $T = 41$.

Step 2: $42 \le T \le 93$. We must show $B(T)/T \ge 65/41$ for every such T. Equivalently, we must show that no fraction B/T with $42 \le T \le 93$ and $B/T > \log_2 3$ satisfies $B/T < 65/41$. Suppose for contradiction that such B, T exist with $\log_2 3 < B/T < 65/41$. Then by [11, Theorem 182], since $65/41 = h_5$ is a principal convergent and $T < q_6 = 53$ (the next convergent denominator) is needed to guarantee that no fraction with smaller denominator beats h_5 on the appropriate side, we observe that $h_6 = 84/53 < \log_2 3$ is a below-approximation, so it does not threaten the above-approximation h_5. The next above-approximation is the upper semiconvergent obtained from the next CF step ($a_7 = 5$ following the $a_6 = 1$ slot), namely the mediant

$$m_1 = \frac{p_5 + 1 \cdot p_6}{q_5 + 1 \cdot q_6} = \frac{65 + 84}{41 + 53} = \frac{149}{94}.$$

By Lemma 3.3, this is the next denominator at which the running minimum of $B(T)/T$ strictly improves; the denominator is 94. Therefore, every T with $42 \le T \le 93$ has $B(T)/T \ge 65/41$,

and equality occurs only at $T = 82$ (twice the denominator 41, as in Remark 3.4); this is verified by direct calculation:

$$B(82) \;=\; \lfloor 82 \cdot \log_2 3 \rfloor + 1 \;=\; \lfloor 129.9669\ldots \rfloor + 1 \;=\; 130, \qquad 130/82 = 65/41.$$

Step 3: Sharpness at $T = 94$. Direct computation gives $94 \log_2 3 = 148.9866\ldots$, so $B(94) = 149$ and $B(94)/94 = 149/94 = 1.585106\ldots < 65/41$. Thus $T = 94$ is the smallest positive integer T for which $B(T)/T < 65/41$.

Combining Steps 1–3 yields the theorem. □

Remark 3.5 (Why h_5 and not h_3 or h_7). The reader may ask why the sixth convergent $h_5 = 65/41$, rather than $h_3 = 8/5$ or $h_7 = 485/306$, is the empirically observed value in Section 4. The answer is one of scale. At $T = 5$, the bound $B(T)/T \geq 8/5 = 1.6$ is comfortably above $\log_2 3 = 1.585$; orbits of length 5 have descent ratios significantly above the worst-case asymptote. By contrast, by the time T reaches 41, the deterministic minimum $B(T)/T$ has descended to $65/41 = 1.5854$, only 0.04% above $\log_2 3$; this is essentially the asymptotic value, and orbits of length ≥ 41 rarely achieve a substantially better ratio (the next improvement only kicks in at $T = 94$ with $149/94$, a relative improvement of just 0.016%). Section 4 shows that typical worst-case Collatz orbits in the bit-length range we sampled hit lengths T that are most often around 41, 82, or small multiples thereof, locking in the $65/41$ value.

Remark 3.6 (Length of validity). Theorem 3.1 holds for $T \leq 93$. Beyond this range the deterministic minimum drops further: from $T = 94$ to $T = 146$ the bound is $149/94$, then $233/147$, then $317/200$, and so on, each slightly closer to $\log_2 3$ from above. The next principal convergent above $\log_2 3$ is $h_7 = 485/306$, so this descent proceeds in small steps. The empirical study in Section 4 works at scales (orbit lengths T up to a few hundred for $k \leq 44$) where these tighter bounds are theoretically allowed but, as we shall see, only $149/94$ ever *occurs* in our data, never $233/147$ or anything beyond.

4 Computational Experiments

This section describes the computational protocol used to investigate worst-case descent ratios of compressed Collatz orbits and reports the results. The code is publicly available at `https://github.com/weiqi-kids/collatz-research-report`.

4.1 Setup and definitions

Definition 4.1 (Length of first descent)**.** For an odd integer $n \geq 3$, the *length of first descent* $T_{\mathrm{des}}(n)$ is the smallest $T \geq 1$ such that $S^T(n) < n$, when such a T exists. We set $T_{\mathrm{des}}(n) = \infty$ otherwise.

By Lemma 2.2, for every $n \geq 3$ that eventually descends below itself (which, by exhaustive verification of the Collatz conjecture for $n \leq 2^{68}$ [2], includes every n in our search range), the length $T_{\mathrm{des}}(n)$ is finite. We then define

$$B_{\mathrm{des}}(n) \;:=\; B_{T_{\mathrm{des}}(n)}(n), \qquad \rho_{\mathrm{des}}(n) \;:=\; B_{\mathrm{des}}(n)/T_{\mathrm{des}}(n).$$

Definition 4.2 (Worst-case descent ratio in a bit-bin)**.** For $k \geq 2$, let $\Omega_k = \mathbb{N}_{\mathrm{odd}} \cap [3, 2^k - 1]$ denote the odd integers with at most k bits. The *worst-case descent ratio* in bit-bin k is

$$\rho^*(k) \;:=\; \min\bigl\{\rho_{\mathrm{des}}(n) : n \in \Omega_k\bigr\}.$$

By Theorem 3.1 (and the trivial extension to $T \geq 94$), $\rho^*(k) > \log_2 3$ holds for every finite k.

Algorithm 1. `first_descent_ratio`(n).

Input: odd integer $n \geq 3$.
Output: pair $(T_{\text{des}}, B_{\text{des}})$ such that $S^{T_{\text{des}}}(n) < n$ and T_{des} is minimal with this property.

1. $m \leftarrow n;\ T \leftarrow 0;\ B \leftarrow 0.$
2. **while** $m \geq n$ **do**
   2a. $u \leftarrow 3m + 1;\ v \leftarrow \mathrm{v}_2(u).$
   2b. $m \leftarrow u/2^v;\ T \leftarrow T + 1;\ B \leftarrow B + v.$
3. **return** (T, B).

4.2 Algorithm

We use the simple Algorithm 4.2 to compute $T_{\text{des}}(n)$, $B_{\text{des}}(n)$, and the corresponding descent ratio.

The algorithm terminates for every n in our search range, since the compressed Collatz iterate has been verified down to 1 for all $n \leq 2^{68}$ [2], and a T-step descent below n occurs in a finite prefix of the orbit.

4.3 Experimental design

We carry out three increasingly large experiments.

Experiment E1 (exhaustive enumeration). For each $k \in \{2, 3, \ldots, 18\}$, we compute $T_{\text{des}}(n)$, $B_{\text{des}}(n)$ and $\rho_{\text{des}}(n)$ for every $n \in \Omega_k$, and record $\rho^*(k)$ as well as the integer pair (T, B) that realizes it. The total work is on the order of $2^{18} \approx 2.6 \times 10^5$ descent computations, each running in microseconds; total runtime is a few seconds.

Experiment E2 (Mersenne and trailing-1 orbits). For each $k \in \{5, 10, 15, 20, 25, 30, 35, 40, 44\}$, we examine the Mersenne orbit starting at $n = 2^k - 1$ and the orbits starting at $n = 2^k m' - 1$ for small odd m'. These are theoretically known to have long initial bad-step segments (by the closed-form shift identity $S^{t-1}(2^t m' - 1) = 2 \cdot 3^{t-1} m' - 1$ from [5], which we use as a black box) and are therefore candidate sources of worst-case descent ratios.

Experiment E3 (structured sampling for $19 \leq k \leq 44$). For each k in this range, we compute $\rho_{\text{des}}(n)$ for a mixed sample drawn from four sources, totaling between 10^4 and 10^5 samples per k:

(a) the Mersenne number $2^k - 1$ and the integers $2^k m' - 1$ for odd $m' \in \{1, 3, 5, \ldots, 999\}$ (*trailing*-1 *patterns*, contributing the long bad-step segments);

(b) the integer that achieved $\rho^*(k - 1)$ in the previous bit-bin together with its small odd multiples (*previous-round expansion*);

(c) 10^4 pseudo-random odd integers drawn uniformly from $\Omega_k \setminus \Omega_{k-1}$ (*random*);

(d) all integers congruent to -1 modulo 2^{k-6} within Ω_k (*near-Mersenne mosaic*).

This sampling strategy is designed to capture orbits that empirically produce small descent ratios, complementing the exhaustive enumeration at lower k. We do not claim that the sample exhausts all worst-case candidates; this is precisely the empirical character of the observation discussed in Section 5.

Environment. The experiments were run on a single workstation (Intel x86-64, `Python 3.11`, native big-integer arithmetic). Total wall-clock time is dominated by E3 and amounts to under one CPU-hour.

4.4 Results

Experiment E1. Table 2 reports $\rho^*(k)$ for $k = 5, \ldots, 18$ in fully-enumerated form. The realizing (T, B) pair is given as a fraction B/T in lowest terms, alongside the matching continued-fraction object.

k	$\#\Omega_k$	worst-case n	B/T ratio	continued-fraction interpretation
5	16	27	59/37	non-canonical (small-T regime)
6	32	63	$54/34 = 27/17$	upper semiconvergent below h_3
7	64	97	27/17	upper semiconvergent
8	128	255	46/29	upper semiconvergent
9	256	447	65/41	h_5
10	512	703	65/41	h_5
11	1024	1639	65/41	h_5
12	2048	1819	65/41	h_5
13	4096	4591	65/41	h_5
14	8192	8255	27/17	upper semiconvergent (anomaly, see text)
15	16384	25295	65/41	h_5
16	32768	52527	65/41	h_5
17	65536	93823	65/41	h_5
18	131072	154079	65/41	h_5

Table 2: Experiment E1: exhaustive worst-case descent ratios $\rho^*(k)$ over $\Omega_k = \mathbb{N}_{\text{odd}} \cap [3, 2^k - 1]$ for $5 \leq k \leq 18$. "h_5" denotes the sixth principal convergent 65/41 of $\log_2 3$; "upper semiconvergent" denotes a best-above approximation between two principal convergents. The $k = 14$ entry is an anomaly arising because the first descent of $n = 8255 = 2^{13} + 63$ has $T_{\text{des}} = 17$, a denominator at which the deterministic minimum is exactly 27/17, not yet 65/41; once k grows enough for the worst-case orbit to require $T_{\text{des}} \geq 29$, the value transitions to 46/29 and ultimately 65/41.

Experiment E2. For Mersenne starting points $n = 2^k - 1$ we observed (with the notation of Definition 2.1):

k	$n = 2^k - 1$	T_{des}	B_{des}	ratio ρ_{des}
5	31	4	7	1.7500
10	1023	9	19	2.1111
15	32767	14	27	1.9286
20	1048575	19	36	1.8947
25	33554431	24	42	1.7500
30	$2^{30} - 1$	29	46	1.5862
41	$2^{41} - 1$	41	65	$1.5854 = 65/41$
44	$2^{44} - 1$	43	69	1.6047

Table 3: Experiment E2: descent statistics of Mersenne starting points. All Mersenne orbits we tested descend within $T = k$ steps or fewer, and at $k = 41$ the descent ratio achieves the continued-fraction value exactly.

The Mersenne orbit at $k = 41$ is particularly noteworthy: its length-41 descent achieves the value 65/41 *exactly*, saturating the bound of Theorem 3.1.

Experiment E3. For $19 \le k \le 44$ we report the worst-case observed descent ratio $\rho^*_{\text{obs}}(k)$ in Table 4.

bit-bin k	$\rho^*_{\text{obs}}(k)$	matches
19	149/94	next semiconvergent
20 to 28	65/41 in every k	h_5
29	149/94	next semiconvergent
30	65/41	h_5
31	149/94	next semiconvergent
32 to 44	65/41 in every k	h_5

Table 4: Experiment E3: worst-case observed descent ratios in bit-bins 19 through 44 over the structured sample of $\sim 10^4$ to 10^5 orbits per bin. All observed values lie in $\{65/41, 149/94\}$, the only two "upper" fractions of $\log_2 3$ with denominator at most 94 and value below 1.59. No tested orbit achieved a ratio strictly between $\log_2 3$ and 149/94.

Distributional remarks. For the worst-case orbit at $k = 18$ ($n = 154{,}079$, with $T_{\text{des}} = 41$ and $B_{\text{des}} = 65$), the distribution of valuations along the descent is

$$\#\{i : \mathrm{v}_i = 1\} = 28, \quad \#\{i : \mathrm{v}_i = 2\} = 5, \quad \#\{i : \mathrm{v}_i = 3\} = 7, \quad \#\{i : \mathrm{v}_i = 6\} = 1. \tag{6}$$

Hence $B = 28 + 10 + 21 + 6 = 65$ and $T = 41$, so $\rho = 65/41$ exactly. We emphasize that this distribution is heavily skewed toward $\mathrm{v} = 1$ (the minimum possible value), and a single occurrence of $\mathrm{v} = 6$ in the segment is what allows the ratio to remain at the boundary value 65/41.

5 The 65/41 Empirical Conjecture

We now formalize the empirical observation of Section 4 into an explicit conjecture. We emphasize that this is a *conjecture*, not a theorem: the supporting evidence is computational and confined to bit-lengths $k \le 44$.

Conjecture 5.1 (The 65/41 barrier conjecture)**.** For every odd integer $n \ge 3$, the descent ratio of n under the compressed Collatz map satisfies

$$\rho_{\text{des}}(n) \;=\; \frac{B_{\text{des}}(n)}{T_{\text{des}}(n)} \;\ge\; \frac{65}{41}. \tag{7}$$

Equivalently, every length-T initial segment of a compressed Collatz orbit at which a first descent occurs satisfies $B_T/T \ge 65/41$.

Remark 5.2. Conjecture 5.1 as stated is *stronger* than what Theorem 3.1 alone would predict. Theorem 3.1 establishes the deterministic bound $B(T)/T \ge 65/41$ for $T \le 93$, but *not* for $T \ge 94$, where the deterministic minimum drops to 149/94. The conjecture asserts that despite this, no orbit of S *actually realizes* a descent ratio strictly between $\log_2 3$ and 65/41, even at lengths $T \ge 94$ where the deterministic floor permits it. Geometrically: while $149/94 \approx 1.5851$ is the lowest *algebraically possible* ratio for $T = 94$, no Collatz orbit appears to ever *achieve* this lower value as a first-descent ratio. The phenomenon thus combines a deterministic continued-fraction bound (which is tight at $T = 41$) with a separate empirical phenomenon (orbits don't reach denominators much larger than 41 in their first descents at the sampled scales).

Remark 5.3. A weaker form of Conjecture 5.1 that may be more tractable is:

$$\liminf_{n\to\infty} \rho_{\mathrm{des}}(n) \;\geq\; \log_2 3.$$

This weaker form is implied by the Collatz conjecture itself (every orbit eventually reaches 1, hence the average valuation exceeds $\log_2 3$ over sufficiently long segments) but is not known unconditionally; Tao's almost-everywhere bound [17] gives the inequality almost surely with respect to logarithmic density. The strong form $\geq 65/41$ in Conjecture 5.1 is what the data of Section 4 actually support.

Observation 5.4 (Worst-case orbit shape)**.** In every worst-case orbit identified in Section 4, the length of first descent T_{des} takes one of the values $T_{\mathrm{des}} \in \{41, 82, 94\}$. At $T = 41$ and $T = 82$ the realized ratio is $65/41$; at $T = 94$ the realized ratio is $149/94$. No worst-case orbit we identified had T_{des} in the gap $(41, 82) \setminus \{41, 82\}$ or strictly greater than 94 within our sampled range.

Remark 5.5 (Connection to length statistics). Observation 5.4 relates Conjecture 5.1 to the statistics of T_{des}. If a future result established that for sufficiently large n, the length $T_{\mathrm{des}}(n)$ either lies in $\{1, 2, \ldots, 93\}$ or grows large enough that the empirical valuation mean concentrates above $\log_2 3$ (in a quantitative sense), then Conjecture 5.1 would follow. The first regime is the regime governed by Theorem 3.1; the second regime is governed by the equidistribution of orbits modulo small powers of 2, which is the content of the questions left open by [17].

6 Cycle-Element Upper Bound

We close with a rigorous corollary of the basic descent equation combined with the transcendence of $\log_2 3$. This section is logically independent of the experimental Sections 4 and 5.

6.1 The cycle equation

Definition 6.1 (Cycle of S)**.** A *cycle* of the compressed Collatz map of period $P \geq 1$ is a finite sequence $(m_0, m_1, \ldots, m_{P-1})$ of odd integers with $m_{i+1} = S(m_i)$ for $0 \leq i \leq P-2$ and $S(m_{P-1}) = m_0$. The cycle is *nontrivial* if $\{m_0, \ldots, m_{P-1}\} \neq \{1\}$ (the trivial cycle is the fixed point $S(1) = 1$ of period $P = 1$).

Lemma 6.2 (Cycle equation)**.** *Let $(m_0, m_1, \ldots, m_{P-1})$ be a cycle of S of period P, let $\mathrm{v}_i = \mathrm{v}_2(3m_{i-1} + 1)$ for $1 \leq i \leq P$, and let $B = \sum_{i=1}^{P} \mathrm{v}_i$. Then*

$$m_0\big(2^B - 3^P\big) \;=\; \sum_{j=0}^{P-1} 3^{P-1-j}\, 2^{B-B_{j+1}}, \tag{8}$$

where $B_{j+1} = \sum_{i=1}^{j+1} \mathrm{v}_i$. In particular, the right-hand side is a positive integer.

Proof. Apply Lemma 2.2 with $T = P$ to the orbit starting at m_0. The condition $S^P(m_0) = m_0$ becomes $(3^P m_0 + C_P)/2^B = m_0$, i.e. $m_0(2^B - 3^P) = C_P$, with $C_P = \sum_{j=0}^{P-1} 3^{P-1-j} 2^{B-B_{j+1}}$ as in Lemma 2.2. Since each term $3^{P-1-j} 2^{B-B_{j+1}}$ is a positive integer (the exponent $B - B_{j+1} \geq 0$), the sum is a positive integer. □

6.2 Transcendence input

Lemma 6.3 (Gelfond–Schneider for $\log_2 3$)**.** *For all positive integers B and P, $2^B \neq 3^P$.*

Proof. If $2^B = 3^P$ for some positive integers B, P, then $B \log 2 = P \log 3$, so $\log_2 3 = B/P$, a rational number. By the Gelfond–Schneider theorem [10, 15], the number $\log_2 3 = \log 3/\log 2$ is transcendental (it is the exponent in $2^{\log_2 3} = 3$, with 2 algebraic and not equal to $0, 1$, and $\log_2 3$ irrational), in particular irrational, so $B/P = \log_2 3$ is impossible. Thus $2^B \neq 3^P$. □

Remark 6.4. A more elementary version of Lemma 6.3 suffices for the purposes of Corollary 6.5: the irrationality of $\log_2 3$, which is provable from unique prime factorization ($2^B = 3^P$ would express the integer 2^B both with and without the prime 3, a contradiction unless $B = P = 0$). The full Gelfond–Schneider theorem is needed only for sharper statements about $|2^B - 3^P|$, which we do not pursue here.

6.3 The cycle-element bound

Corollary 6.5 (Cycle-element upper bound)**.** *Let $(m_0, m_1, \ldots, m_{P-1})$ be a nontrivial cycle of the compressed Collatz map of period P with cumulative valuation $B = \sum_{i=1}^{P} \mathrm{v}_i$. Then $2^B - 3^P \geq 1$, and the smallest element $m_{\min} = \min_i m_i$ of the cycle satisfies*

$$m_{\min} \;<\; \frac{2^B \cdot 3^P}{2\,(2^B - 3^P)}. \tag{9}$$

In particular, $B > P \log_2 3$, so $B/P > \log_2 3$ for every nontrivial cycle.

Proof. Existence of a cycle implies existence of $m_0 > 0$ satisfying (8). The right-hand side of (8) is positive and $m_0 > 0$, so $2^B - 3^P > 0$. By Lemma 6.3 the quantities 2^B and 3^P are distinct positive integers, so $2^B - 3^P \geq 1$.

To bound m_0 from above, observe from (8) that

$$\begin{aligned} m_0(2^B - 3^P) = \sum_{j=0}^{P-1} 3^{P-1-j}\, 2^{B-B_{j+1}} \;&\leq\; 2^B \sum_{j=0}^{P-1} 3^{P-1-j} \;=\; 2^B \cdot \frac{3^P - 1}{3 - 1} \\ &< \frac{2^B \cdot 3^P}{2}. \end{aligned}$$

Dividing by $2^B - 3^P \geq 1$ yields $m_0 < 2^B \cdot 3^P/(2(2^B - 3^P))$.

The same argument applies starting from any cyclic rotation of the sequence (changing the starting point m_0 to m_i); choosing i such that $m_i = m_{\min}$ proves the bound for the smallest element.

Finally, $2^B - 3^P \geq 1 > 0$ gives $2^B > 3^P$, so $B > P \log_2 3$, i.e. $B/P > \log_2 3$. □

Remark 6.6 (Numerical instance). Setting $(B, P) = (65, 41)$ in Corollary 6.5 gives

$$2^{65} - 3^{41} \;=\; 36{,}893{,}488{,}147{,}419{,}103{,}232 - 36{,}472{,}996{,}377{,}170{,}786{,}403 \;=\; 420{,}491{,}770{,}248{,}316{,}829,$$

hence $m_{\min} < 2^{65} \cdot 3^{41}/(2 \cdot 420{,}491{,}770{,}248{,}316{,}829) \approx 1.6 \times 10^{21}$. Since the Collatz conjecture has been verified for $n \leq 2^{68} \approx 2.95 \times 10^{20}$ [2], the bound (9) for $(B, P) = (65, 41)$ is not strong enough to rule out a cycle of this (B, P) shape unconditionally. This is consistent with Eliahou's lower bound for cycle periods [9], which forces $P \geq 17$ but allows much larger cycles.

Remark 6.7 (Why the bound is not enough). The bound in Corollary 6.5 grows roughly as $3^P/P$ (for $B \approx P \log_2 3$, the denominator $2^B - 3^P$ can be much smaller than 3^P). Combined with Eliahou's lower bound on P, the resulting bound on $m_{\min}$ is enormous, far exceeding the verified range [2]. Stronger bounds require a *linear forms in logarithms* approach (Baker–style theorems; see [1]) on $|B \log 2 - P \log 3|$, which give quantitative lower bounds for $|2^B - 3^P|$ (and hence upper bounds on $m_{\min}$) that decay only polynomially in P, still leaving a large gap. Closing this gap is one of the standard open problems in the cycle-elimination literature; see [16] for a classical attack restricted to "1-cycles".

7 Discussion

The role of 65/41 in the descent ratio landscape. Theorem 3.1 pinpoints 65/41 as the deterministic minimum of $B(T)/T$ in the range $T \leq 93$, and Conjecture 5.1 asserts the (much stronger) empirical fact that no first-descent orbit segment of any length attains a ratio strictly between $\log_2 3$ and 65/41. The first statement is a fact about the continued fractions of $\log_2 3$; the second statement involves specific Collatz orbits and is, at the present level of theory, inaccessible.

What the empirical match does and does not establish. The empirical match between worst-case observed descent ratios and the deterministic continued-fraction minimum at $T \leq 93$ is striking but constitutes *evidence*, not proof. Specifically, the data of Section 4 establish that for every odd $n < 2^{18}$, and for every n in our structured sample for $2^{18} \leq n < 2^{44}$, the inequality $\rho_{\text{des}}(n) \geq 65/41$ holds. They do not establish that this inequality persists for all n. Two cautionary observations are in order.

First, even granting Conjecture 5.1, this would not directly imply the Collatz conjecture: a divergent orbit could in principle exist while its first-descent segments still have ratio $\geq 65/41$ (the orbit would simply contain only finitely many descents below the starting value). The conjecture rules out descent ratios approaching $\log_2 3$ from above too quickly, but it does not rule out orbits that fail to descend at all. Equivalently, Conjecture 5.1 is a quantitative refinement of the "Lemma D" descent statement (every orbit has *some* descent window) but does not prove that statement; the latter remains equivalent to the Collatz conjecture itself.

Second, the empirical sample for $19 \leq k \leq 44$ in Experiment E3 is structured but not exhaustive. The structured sample is designed to capture orbits that should produce small descent ratios (trailing-1 patterns, near-Mersenne, expansions of previously worst-case orbits), and the consistency of the result over 25 distinct bit-bins is evidence in favor of the conjecture. A genuinely adversarial integer outside the sample could in principle violate it.

Connection to the Diophantine literature. The cycle-element bound Corollary 6.5 reproduces, in the notation of the compressed Collatz map, a classical observation of Crandall [8] and Eliahou [9]. Our contribution here is mostly expositional: the bound comes out very cleanly from the closed-form $S^P(m_0) = m_0$ equation (Lemma 6.2) combined with Gelfond–Schneider (Lemma 6.3), which we record because the same machinery is the natural rigorous companion to the experimental Sections 4–5.

The numerical strength of the cycle bound is governed by lower bounds on $|2^B - 3^P|$. The continued-fraction expansion of $\log_2 3$ shows that $|2^B - 3^P|$ can be as small as $3^P/(\text{integer})$ in worst case (when (B, P) is a convergent denominator pair), corresponding precisely to the same convergents that drive Theorem 3.1. This is the same arithmetic phenomenon viewed from two angles.

Open questions. Several questions naturally suggest themselves.

(1) *Quantitative version of Conjecture 5.1.* Can one give an effective bound, in terms of n, on the smallest descent ratio $\rho_{\text{des}}(n)$ that exceeds $\log_2 3$? The data suggest the bound is 65/41 uniformly, but even a slightly weaker bound (e.g. $\rho_{\text{des}}(n) \geq \log_2 3 + c/\log n$ for some explicit $c > 0$) would be of interest.

(2) *Length statistics of first descents.* What is the distribution of $T_{\text{des}}(n)$ for random odd n of bit-length k? The current data suggest a heavy concentration near $T \in \{41, 82\}$, but a precise asymptotic is unknown.

(3) *Further continued-fraction phenomena.* Theorem 3.1 gives the worst case in $T \leq 93$. Does a phase transition occur at, say, $T = q_7 = 306$ (the seventh convergent denominator),

corresponding to the next jump in the running minimum of $B(T)/T$? Computational extension to $k > 44$ would be needed to test this.

(4) *Sharpening the cycle bound.* A Baker-type lower bound for $|B \log 2 - P \log 3|$ specialized to the case where (B, P) is forced to obey Conjecture 5.1 (i.e. $B/P \geq 65/41$) might give sharper bounds than Corollary 6.5 alone.

Reproducibility. All code used to generate the tables in Section 4 is available in the repository `https://github.com/weiqi-kids/collatz-research-report`, specifically in the files `collatz_experiment.py` and `collatz_deep_analysis.py`. The reported running times were measured on a single workstation; total CPU usage across all experiments is under one CPU-hour. Verification of the small-T cases of Theorem 3.1 (for example, the claim that $B(T)/T \geq 65/41$ for $T \leq 93$) reduces to computing $B(T) = \lfloor T \log_2 3 \rfloor + 1$ for 93 values of T and comparing to $65/41$; this can be done by hand or in any computer algebra system.

8 Conclusion

We have isolated a precise interaction between the continued-fraction expansion of $\log_2 3$ and the descent dynamics of the compressed Collatz map. The deterministic minimum descent ratio $B(T)/T$ for $T \leq 93$ is governed by the sixth principal convergent $h_5 = 65/41$ (Theorem 3.1); an extensive computational study supports the strong conjecture that this ratio is the actual worst case across all Collatz orbits at sampled scales (Conjecture 5.1); and the same arithmetic input (transcendence of $\log_2 3$) yields a clean cycle-element upper bound (Corollary 6.5). None of these results resolves the Collatz conjecture, but together they pinpoint the exact arithmetic obstruction that any future quantitative theory will need to address.

References

[1] A. Baker, *Transcendental Number Theory*, 2nd ed., Cambridge University Press, Cambridge, 1990.

[2] D. Barina, *Convergence verification of the Collatz problem*, J. Supercomput. **77** (2021), 2681–2688.

[3] D. J. Bernstein and J. C. Lagarias, *The $3x + 1$ conjugacy map*, Canad. J. Math. **48** (1996), 1154–1169.

[4] Y. Bugeaud, *Approximation by Algebraic Numbers*, Cambridge Tracts in Mathematics, vol. 160, Cambridge University Press, Cambridge, 2004.

[5] L. Chang, *A three-state finite automaton for carry propagation in the accelerated Collatz map: spectral gap and closed-form shift dynamics*, preprint, 2026.

[6] L. Chang, *No finite local potential function is a strict Lyapunov function for the accelerated Collatz map: an impossibility result*, preprint, 2026.

[7] J. H. Conway, *Unpredictable iterations*, in: Proc. 1972 Number Theory Conference, Univ. of Colorado, 1972, pp. 49–52.

[8] R. E. Crandall, *On the "$3x + 1$" problem*, Math. Comp. **32** (1978), 1281–1292.

[9] S. Eliahou, *The $3x + 1$ problem: new lower bounds on nontrivial cycle lengths*, Discrete Math. **118** (1993), 45–56.

[10] A. Gelfond, *Sur le septième problème de Hilbert*, Bull. Acad. Sci. URSS **7** (1934), 623–634.

[11] G. H. Hardy and E. M. Wright, *An Introduction to the Theory of Numbers*, 6th ed., Oxford University Press, Oxford, 2008.

[12] A. Y. Khinchin, *Continued Fractions*, University of Chicago Press, Chicago, 1964.

[13] J. C. Lagarias, *The* $3x+1$ *problem and its generalizations*, Amer. Math. Monthly **92** (1985), 3–23.

[14] J. C. Lagarias (ed.), *The Ultimate Challenge: The* $3x+1$ *Problem*, American Mathematical Society, Providence, RI, 2010.

[15] T. Schneider, *Transzendenzuntersuchungen periodischer Funktionen, I: Transzendenz von Potenzen*, J. reine angew. Math. **172** (1934), 65–69.

[16] R. P. Steiner, *A theorem on the syracuse problem*, in: Proc. 7th Manitoba Conf. on Numerical Math. and Computing, Congress. Numer. **20**, Utilitas Math., Winnipeg, 1977, pp. 553–559.

[17] T. Tao, *Almost all orbits of the Collatz map attain almost bounded values*, Forum Math. Pi **10** (2022), e12.

[18] G. J. Wirsching, *The Dynamical System Generated by the* $3n+1$ *Function*, Lecture Notes in Math. **1681**, Springer, Berlin, 1998.